NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED **MATHEMATICS** **M.M. 73**

REVISION ASSINGMENT # 13 (INDEFINITE AND DEFINITE INTEGRATION-01) TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If $f(x)$ is an even function which is also periodic with the period T and $\int_0^a f(x)dx = 3$ and $\int_{-T/2}^{3T/2} f(x)dx = 18$, then $\int_{-a}^{a+5T} f(x)dx$ is equal to -

(A) 96 (B) 93 (C) 51 (D) 48

2. The value of $\int_{3/2}^{15} [\ell n[x]]dx$ is equal to, (where [.] denotes greatest integer function) -

(A) $\frac{19}{2}$ (B) 9 (C) 19 (D) $\frac{9}{2}$

3. Let for every integer a, $\int_a^{a+1} f(x)dx = a^2$, then the value of $\int_{-2}^{4} f(x)dx$ is equal to -

(A) 16 (B) 17 (C) 18 (D) 19

4. The value of definite integral $\int_{-1}^{1} \frac{x\tan^{-1}x}{1+e^{\tan^{-1}x}}dx$ is :-

(A) $\frac{\pi}{4} - \ell n2$ (B) $\frac{\pi}{4} - \frac{1}{2}$ (C) $\frac{\pi}{4} + \frac{1}{2}$ (D) $\frac{\pi}{2} - 1$

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5. $\int \frac{x^2-4x+6}{x^2-2x+4}dx$ is equal to -

(A) $x + \ell n(x^2 - 2x + 4) + c$ (B) $\ell n(x^2 - 2x + 4) + 2x + c$

(C) $2x - \ell n(x^2 - 2x + 4) + c$ (D) $x - \ell n(x^2 - 2x + 4) + c$

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6. $\int \frac{(1-\tan^2 x).\text{cosec}2x\,dx}{\sec^2 x}$ is equal to

(A) $\frac{\ell n|\sin 2x|}{2}+C$ (B) $\frac{\ell n|\cos 2x|}{2}+C$ (C) $\frac{\ell n(\tan x)}{2}+C$ (D) $\frac{\ell n|\cos x+\sin x|}{2}+C$

7. If $\int e^x\left(\ell nx + x\ell nx + 1\right)dx = f(x) + C$ when $f(1) = 0$, then $f(e)$ is equal to-

(A) e (B) e^e (C) e^{e-1} (D) e^{e+1}

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

8. If $I_{(n)} = \int_0^{\pi}(1 - |\sin x| + |\sin 2x| - |\sin 3x| + \ldots\ldots$ upto n terms) dx then -

(A) $I_{(2)} = \pi - 2$ (B) $I_{(3)} = \pi$ (C) $I_{(2008)} = \pi - 2$ (D) $I_{(2009)} = \pi$

9. If $I_n = \int_0^1 (1-x^2)^n dx$, then -

(A) $I_n = \frac{2n}{2n+1}I_{n-1}$ (B) $I_n = \frac{2.4.6.\ldots\ldots.2n}{3.5.7.\ldots\ldots.(2n+1)}$ (C) $I_n = \frac{2^n n!}{3.5.7.\ldots\ldots.(2n+1)}$ (D) $I_n = \frac{2n+1}{2n}I_{n-1}$

10. $\int \frac{3e^x + 5e^{-x}}{4e^x - 5e^{-x}}dx = Ax + B\ell n|4e^{2x} - 5| + C$ then -

(A) A is –1 (B) B is $\frac{7}{8}$ (C) C can be –1 (D) A = 1 & $B = -\frac{7}{8}$

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11 Let $f : [0,\infty) \to R$ be a continuous strictly increasing function, such that $f^3(x) = \int_0^x t f^2(t)dt$ $\forall\ x \geq 0$, then which of the following is/are **INCORRECT-**

(A) $f(x)$ is onto function

(B) $\int_0^1 f(x)dx = \frac{1}{18}$

(C) Number of solutions of $6f(x) = e^x$ is 2

(D) graph of $y = 6f(x)$ intersects the graph of $y = 3x^2 + 2x + 2$ at one point

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SECTION–III

Numerical Grid Type (Single digit Ranging from 0 to 9) (4 Marks each, –1 for wrong answer)

1. Let a differentiable function $f(x)$ satisfies $f(x)\,f'(-x) = f(-x)\,f'(x)$ and $f(0) = 1$, then $\int_{-2}^{2} \frac{dx}{(1+f(x))}$ is equal to

2. The value of integral $\int_{1/2}^{2} e^{\left|x-\frac{1}{x}\right|}dx$ is equal to $\lambda_1\left[e^{\lambda_2/2}-1\right]$ then $\lambda_1 + \lambda_2$ is equal to

3. If $I_1 = \int_0^{100\pi} (\sin(\cos x) + \cos(\sin x))dx$ & $I_2 = \int_0^1 \frac{\cos x dx}{\sqrt{1-x^2}}$, then $\frac{I_1}{50I_2}$ is equal to

4. If $\lim\limits_{n\to\infty} \sum_{r=n}^{2n} \frac{n^4}{r^3\sqrt{r^4+n^4}} = \sqrt{a} - \sqrt{b}$, then $\frac{32b}{a} - 10$ is equal to

5 $\int \left(x^{12}+x^8+x^4\right)\left(2x^8+3x^4+6\right)^{1/4} dx = a\left(2x^{12}+3x^8+6x^4\right)^{5/4} + C$ (where C is integration constant), then the value of $(60a + 1)$ is

SECTION–IV

Matrix Match Type (One or More than one option correct)

For each entry in **Column-I** , +2 If only the bubble(s) corresponding to all the correct matche(es) is (are) darkened, 0 In none of the bubbles is darkened, –1 In all other cases

1.

	Column-I		**Column-II**
(A)	$\int_4^{10} \frac{[x^2]dx}{[x^2-28x+196]+[x^2]} =$ {where [.] denotes greatest integer function}	(p)	$\frac{1}{100}$
(B)	$\int_{-1}^{2} \frac{\lvert x\rvert}{x}dx =$	(q)	3
(C)	$\lim\limits_{n\to\infty} \frac{1^{99}+2^{99}+.....+n^{99}}{n^{100}} =$	(r)	$\frac{1}{3}$
(D)	$5050\int_{-1}^{1} \sqrt{x^{200}}dx = \frac{1}{\alpha}$, then $\alpha =$	(s)	1

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2.

	Column-I		**Column-II**
(A)	Let $\int \cos^5 x dx = \sin x + \frac{\sin^3 x}{m} + \frac{\sin^5 x}{n} + c$, then n + 2m is equal to	(P)	2
(B)	Let $\int \frac{xdx}{x^2 - 8x + 15} = \ell n \left\lvert \frac{(x-5)^a}{(x-3)^b} \right\rvert + c$, then a + b is equal to	(Q)	3
(C)	Let y = f(t) & x = g(t) where t is a parameter & f,g are twice derivable functions. If $f'(1) = 0$, g'(1)=1 & $f''(1)=2$ then $\frac{d^2y}{dx^2}$ at t = 1 is equal to	(R)	0
(D)	Let $L = \lim_{x \to 0^+} (-\ell nx)^{\frac{1}{\ell nx}}$, then ℓnL is equal to	(S)	–2
		(T)	Natural number

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REVISION ASSINGMENT # 12 CONTINUITY, DIFFERENTIABILITY & MOD **MATHEMATICS**

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	B	C	B	A	D	A	B	C	D	D
	Q.	11	12	13	14	15	16				
	A.	A,C	A,B,C	B,D	B,C	B	C				
SECTION-III	Q.	1	2	3	4						
	A.	4	0	2	3						
SECTION-IV	Q.1	A	B	C	D						
		R	Q,S	R	T						

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